Image Filtering

Computer Vision / Image Processing Notes

♥ Notes by Humna Imran ♥

Image Filtering Kya Hai? (What is Image Filtering?)

- Simple words mein: Image filtering ek technique hai jisme image ko modify ya enhance kiya jata hai.
- Pixels ki values change ki jati hain taake:
  - Image ko saaf kiya ya sake (noise / daghg-d-dhubbbe remove).
  - Image ki details nikali ja sakein (edges detect).
  - Image ko sharp ya blur kiya ja sake.

Yeh Kaam Kaise Karta Hai? (Core Concept)

- **Original Image**
  - Ek bada matrix hota hai (pixels ke numbers).
- **Kernel / Mask / Filter**
  - Ek chhota matrix hota hai (3×3 ya 5×5).
- **Process – Convolution**
  Kernel image ke upar slide karta hai (left → right, top → bottom).

1	0	-1
1	0	-1
1	0	-1

→ $\sum$ + new value.

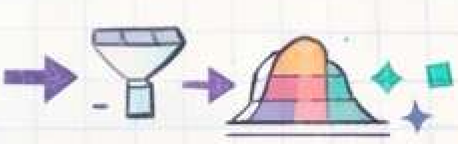
- pixel × kernel values multiply
- sab numbers add
- jo result aaye wo center pixel ki new value ban jata hai.

Image Filters Categories (Exam Important)

Low-Pass Filters (Smoothing / Blurring)

- Yeh filters high frequency changes remove karte hain.
- Main kaam:
  - noise remove karna aur image smooth / blur karna.

1	0	-1
1	0	-1
1	0	-1



- pixel × kernel values multiply
- sab numbers add
- jo result aaye wo center pixel ki new value ban jata hai.

Mean Filter (Average Filter)

Kaam:

- Neighborhood pixels ki average nikal kar center pixel ko assign karta hai.

Fayda:

- Noise kam ho jati hai.

Gaussian Filter

Kaam:

- Center pixel ko zyada weight deta hai aur door pixels ko kam.

Fayda:

- Noise kam ho jati hai.

Exam Tip:

Image filtering mein Kernel, Convolution aur Low-Pass Filters bohat important concepts hain.

Image Filtering – Types of Filters

— Computer Vision Notes —

Notes by Humna Imran.

1. Low-Pass Filters (Smoothing / Blurring Filters)

- Yeh filters image mein se sharp changes (high frequency) ko khatam karte hain.
- Inka main kaam noise remove karna aur image ko smooth / blur karna hota hai.

Mean Filter (Average Filter)

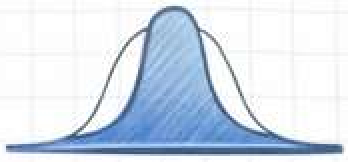
- **Kaam:** Aas-paas ke neighborhood pixels ki average (mean) nikal kar center pixel ko assign karta hai.
- **Fayda:** Image mein se noise kam ho jati hai.
- **Nuqsan:** Edges bhi blur ho jati hain, jis se image thori dhundi liagti hai.

1	1	1
1	1	1
1	1	1

→ 5

Gaussian Filter

- **Kaam:** Center pixel ko zyada weight deta hai aur door pixels ko kam.
- **Fayda:** Mean filter se blur zyada natural hota hai.



- **Nuqsan:** Edges bhi blur ho jati hain, jis se image thori dhundli lagti hai.

1	1	1	1
1	1	1	1

→ 5

2. Non-Linear Filters (The Special One)

- Yeh filters simple multiplication-addition par nahi chalte, balki sorting waghera use karte hain.

Median Filter (Bohat Important)

- **Kaam:** Aas-paas ke pixels ko ascending order mein sort karta hai.
- **Fayda:** se blur zyada natural hota hai.

-1	0	1
-1	0	1
-1	0	1

3. High-Pass Filters (Sharpening & Edge Detection)

- Yeh filters low-frequency areas ko block karte hain aur jahan color achanak change ho (edges / boundaries) unko highlight karte hain.

Edge Detection Filters:

- **Sobel / Prewitt:** Yeh filters image ke horizontal aur vertical edges detect karte hain.

-1	0	1
-1	0	1
1	0	1

Image Filtering Comparison & Image Pyramid

– Computer Vision / Image Processing Notes –

Notes by Humna Imran

Low-Pass vs High-Pass Filters

Feature	Low-Pass Filter	High-Pass Filter
Main Kaam	Image ko blur / smooth karna	Edges dhoondhna / sharpen karna
Frequency	Low frequency pass karta hai	High frequency pass karta hai
Kernel Sum	Kernel ke numbers ka sum usually 1 hota hai	Kernel ke numbers ka sum usually 0 hota hai
Examples	Mean, Gaussian	Sobel, Prewitt, Laplacian

1. Image Pyramid Kya Hai? (What is an Image Pyramid?)

Aasan lafzon mein, Image Pyramid ek hi image ki alag-alag resolutions (sizes) ki collection hoti hai.

Agar hum pyramid (Misar ke Ahram) ko tasawwur karein:

- **Base (Level 0)**
  Sabse badi aur original high-resolution image
- **Top (Highest Level)**
  Sabse choti aur low-resolution image

Jaise jaise hum pyramid mein upar jaate hain, image ka size aur resolution chota hota jata hai.

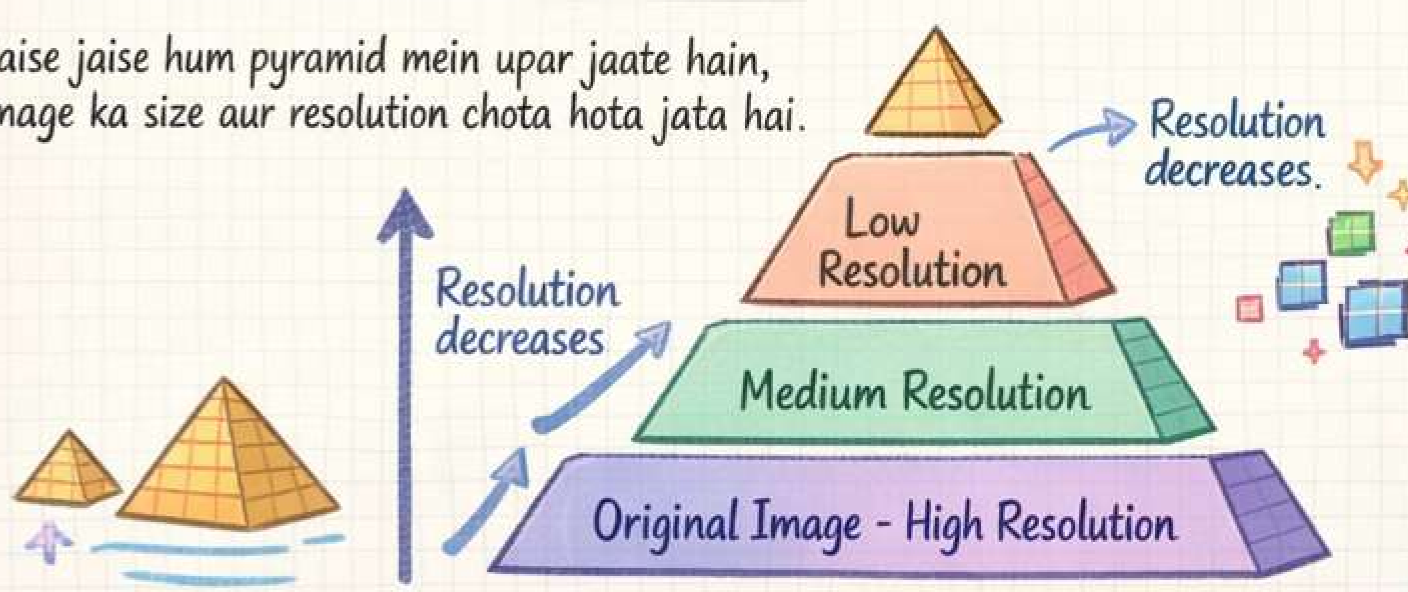


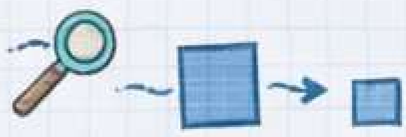
Image Pyramids – Importance & Types

Computer Vision / Image Processing Notes

Notes by Humna Imran

2. Image Pyramids Ki Zaroorat Kyun Hai?

- Computer vision mein humein objects ko detect karna hota hai.
- Masla yeh hai ke real world mein objects different sizes ke ho sakete hain.
- Kuch objects camera ke qareeb (bade) hote hain
  Aur kuch door (chote).



- Ek normal model shayad bade object ko detect kar le lekin chote ko miss kar de.
- Isliye hum image pyramid banate hain taake model har scale par object detect kar sake.

Is concept ko Multi-Scale Representation kehte hain.

3. Types of Image Pyramids

- Exam mein mostly 2 main types poochi jati hain:
  - **Gaussian Pyramid**
  - **Laplacian Pyramid**

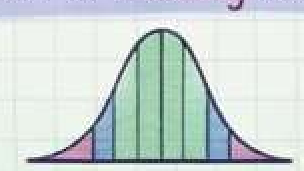
A. Gaussian Pyramid

- Yeh sabse common image pyramid type hai.
- Isme hum base (original image) se start karte hain aur image ko step by-step chota (downsample) karte hain.

Blurring (Smoothing)

- Original image par Gaussian Filter (blur) apply kiya jata hai.
- Iska purpose:
  - High-frequency noise remove karna
  - Sharp edges smooth karna

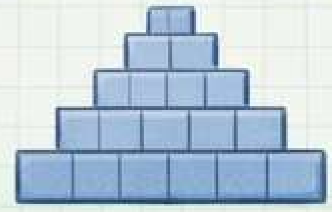
Taake image choti karte waqt informtion karab na ho
Is problem ko Aliasing kehte hain.



Subsampling

- Blur karne ke baad:
  - Image ki even rows aur even columns remove kar di jati hain.
- Result:
  - Image ka size exactly half ho jata hai.

512 → 256 → 128



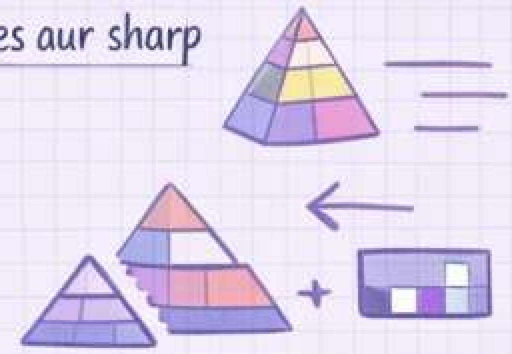
Laplacian Pyramid & Applications

Computer Vision / Image Processing Notes

--- Notes by Humna Imran --

B. Laplacian Pyramid

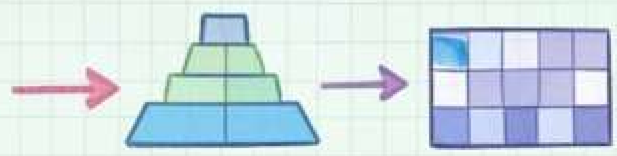
- Laplacian pyramid directly nahi banta, balkay yeh Gaussian Pyramid se banaya jata hai.
- Jab Gaussian pyramid mein image choti ki jati hai, toh edges aur sharp details lost ho jati hain.
- Laplacian pyramid un lost details ko store karta hai.
- Yeh technique image compression aur image reconstruction mein use hoti hai.



Yeh Kaise Kaam Karta Hai? (The Process)

- Step 1 · Upsampling

  Choti image → Badi image.



- Step 2 · Blurring
- Step 3.· Subtraction.

  Original Image
  − Expanded Image

4. Image Pyramids ke Applications (Faide)

- Image Blending.
  - Do alag images ko seamlessly merge karne ke liye.
  - Apple + Orange → "Oraple".

- Object Detection.
  - Face detection ya car detection mein use hota hai.

- Image Compression.
  - Kam storage size mein save karne ke liye.
- Feature Tracking.
  - Video ya optical flow mein object ke movement ko track karna.

Fourier Transform in Image Processing

Computer Vision / Image Processing Notes

- Notes by Humna Imran -

1. Fourier Transform Asal Mein Kya Hai?

- Agar aam zindagi ki misaal lein:

- Agar aap mixed fruit smoothie peete hain, toh Fourier Transform ek aisi machine hai jo bata sakti hai:
  - kitna apple hai
  - kitna banana hai
  - kitni sugar hai

- Computer vision mein:
  - Smoothie = Image
  - Fourier Transform = Mathematical tool.

Yeh tool image ko chote chote components mein break down karta hai jise **Frequencies** kehte hain:

2. Spatial Domain vs Frequency Domain

Spatial Domain

- Yeh normal image representation hoti hai.
- Image pixels (x, y coordinates) se mil kar banti hai.

- Hum apni aankhon se spatial domain hi dekhte

Frequency Domain

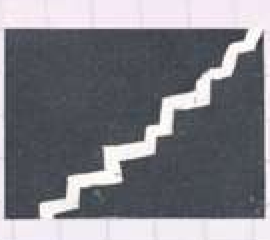
- Jab image par Fourier Transform apply hota hai toh
- Yeh domain batata hai ke image mein pixel values kitni tezi se change ho rahi hain.

3. Image Mein Frequencies Kya Hoti Hain?

Frequency ka matlab hai "Change" (Badlao)

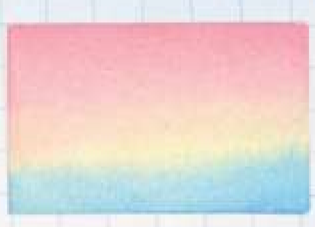
High Frequencies

- Tez Badlao
- Edges / Boundaries
- Sharp details
- Noise



Low Frequencies

- Aahista Badlao
- Smooth surfaces
- Plain background
- Sky / Soft gradients



Fourier Transform – Filtering & Applications

Computer Vision / Image Processing Notes

~ Notes by Humna Imran. ~

4. Frequency Domain Mein Aane Ka Fayda?

Jab image Frequency Domain mein convert ho jati hai, toh hum uske kuch frequencies ko rakh sakte hain aur kuch ko remove kar sakte hain.

Is process ko Filtering kehte hain.

Low Pass Filter (LPF)

- **Yeh kya karta hai?**
  - Sirf Low Frequencies (smooth areas) ko pass karta hai aur High Frequencies (edges / noise) ko block kar deta hai.
- **Result:**
  - Image Blur / Smooth ho jati hai.

- **Use:**
  Noise remove karne ke liye.

High Pass Filter (HPF)

- **Yeh kya karta hai?**
  - Sirf High Frequencies (edges) ko pass karta hai aur Low Frequencies (background) ko block kar deta hai.
- **Result:**
  - Image mein sirf edges aur boundaries highlight ho jati hain.

- **Use:**
  Edge detection aur image sharpening

5. Inverse Fourier Transform (Wapas Aana)

Jab hum Frequency Domain mein filtering kar lete hain, toh humein result ko wapas normal pixel image mein convert karna hota hai.

Is process ko Inverse Fourier Transform (IFT) kehte hain.

6. Computer Vision Mein Applications

- **Image Enhancing / Smoothing**
  Noise remove karna Low Pass Filter ke zariye.
- **Edge Detection**
  Image ke objects ke edges detect karna High Pass Filter ke zariye.

- **Image Compression**
  JPEG images save karte waqt Fourier ya DCT use hota hai.

Hough Transform – Shape Detection

Computer Vision / Image Processing Notes

Notes by Humna Imran

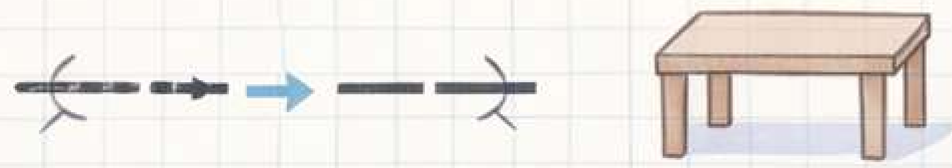
1. Hough Transform Asal Mein Kya Hai?

✔ **Aam zaban mein:**

- Yeh technique computer ko image ke andar shapes detect karne mein madad karti hai.

✔ **Khaas tor par:**

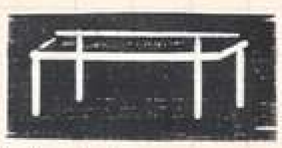
- Straight lines
- Circles
- Ellipses

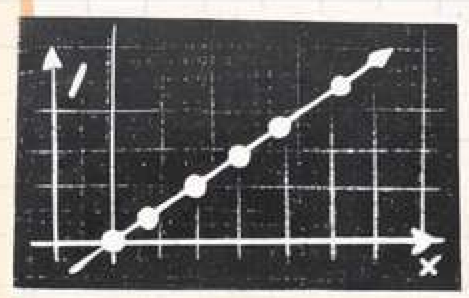


Agar image mein table ki boundary line broken ho (beech se missing ho), toh bhi Hough Transform us line ko detect kar sakta hai.

Exam Point ☆

- Hough Transform direct original image par apply nahi hota.
- Pehle image par Edge Detection ki jati hai. (jese Canny Edge Detector).
- Uske baad edge image par Hough Transform apply hota hai.





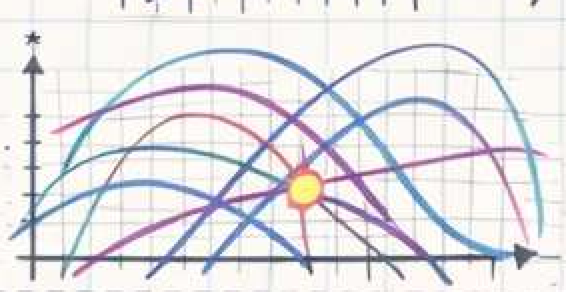
2. Image Space vs Hough Space

The Core Logic

✔ Is concept mein do different spaces hote hain.

- **Image Space (x, y)**
  - Yeh normal image domain hota hai.
  - Ek straight line bohot saare pixels (points) se mil kar banti hai.

- **Hough Space (Parameter Space)**
  - Yeh mathematical parameter space hota hai.
  - Image space ka har point (x, y) Hough space mein ek curve ban jata hai.



Key Concept Diagram

Intersection = Detected Line

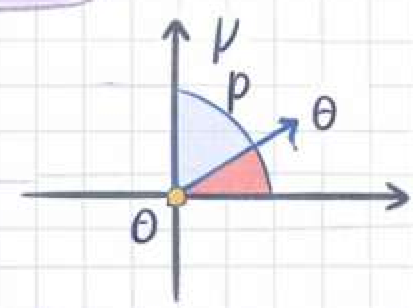
Hough Transform – Polar Coordinates & Voting System

Computer Vision / Image Processing Notes

— Notes by Humna Imran —

3. Cartesian vs Polar Coordinates (Exam Favorite ⭐)

Normally ek line ki equation hoti hai: $y = mx + c$



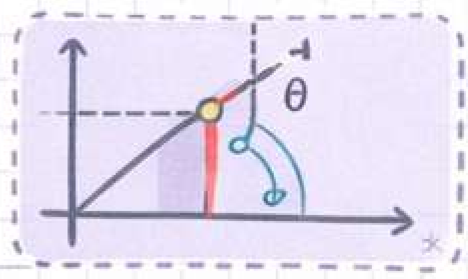
Jahan:

m = slope.

c = intercept

Problem:

- Agar line vertical ho, toh slope infinity (∞) ho jata hai.
- Computer infinity ko store nahi kar sakta.



Solution – Polar Coordinates

- Polar Coordinates use kiye jate hain: ρ (rho) aur θ (theta)

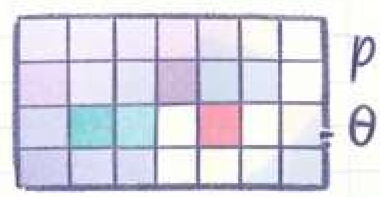
$$\rho = x\cos(\theta) + y\sin(\theta)$$

- ρ (Rho) = origin se line tak ka distance
- θ (Theta) = us distance ka angle.

4. Hough Transform Ka Voting System

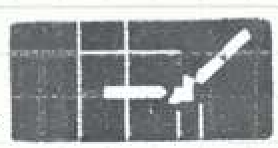
1. Accumulator Grid

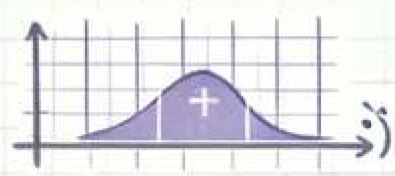
- Computer Hough Space (ρ, θ) ka 2D grid / array banata hai.
- Is grid ko Accumulator kehte hain.



2. Edge Points Check

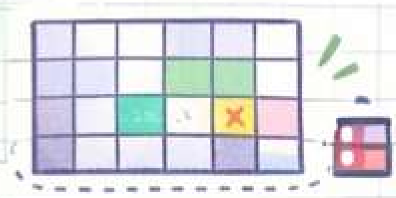
- Image ke har edge pixel ko check kiya jata hai.





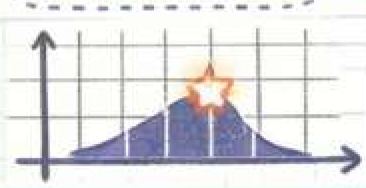
3. Curve Generate

- Formula : $\rho = x\cos(\theta) + y\sin(\theta)$
- Different θ values (0–180°) ke liye ρ calculate kiya jata hai.



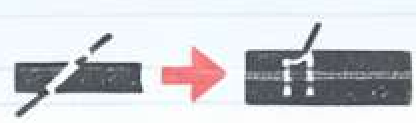
4. Voting

- Jo bhi ρ aur θ values milti hain:
- Accumulator grid ke us cell mein **+1 vote add** hota hai.

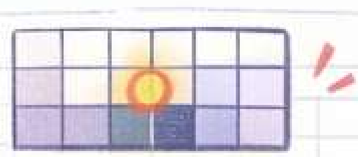


5. Hough Transform ke Fayde

- **Robust to Noise**
  - Image mein noise ho tab bhi lines detect ho jati hain.

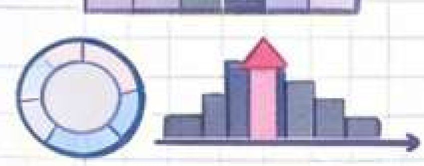


- **Broken Lines Detection**
  - Agar line tooti hui ho ya partially hide ho, phir bhi detect ho sakti hai.



- **Versatile Method**
  - Thori modifiiation se circles aur shapes bhi detect kiye ja sakte hain.

  Example: Hough Circle Transform



Transform Domain in Computer Vision

— Concept Notes —

— Notes by Humna Imran —

1. Transform Domain Asal Mein Kya Hai? (The Core Idea)

- Computer Vision mein image par kaam karne ke do tareeqe hote hain:

- **Spatial Domain**
  - Jab hum direct image ke pixels (x, y) par changes karte hain
  - Example: brightness change, simple blur

- **Transform Domain**
  - Image ko pehle ek mathematical transform ke zariye ek nayi representation mein convert kiya jata hai
  - Wahan processing hoti hai
  - Phir image ko wapas original pixel domain mein laaya jata hai

Aasaan Misaal:
Farz karein aapko English ki book mein se "Aam" (Mango) word dhoondna hai.
Pehle book ko Urdu mein translate (Transform) karte hain,
phir aaraam se "Aam" dhoond kar change karte hain, aur phir wEnglish mein translate kar dete hain.
Transform domain bhi isi tarah kaam karta hai.

2. Processing ka 3-Step Flowchart (Exam ke liye VIP)

Forward Transform

Input image
$f(x,y)$
⇓
$F(u,v)$
Transform Domain representation

→

Processing / Filtering

Transform image ko Filter Mask $H(u, v)$ ke sath multiply kiya jata hai.

$$G(u,v) = F(u,v) \times H(u,v)$$

Yahan important, frequencies ko retain kiya jsta hai aur unnantal frequencies remove is jotr hain.

→

Inverse Transform

Processed transform:
$G(u,v)$
⇓
$g(x, y)$
Jo normal pixel image hoti ha jo hum dekh sakte hain.

Exam Tip:

- Transform Domain processing image filtering aur frequency analysis ke liye bohat important concept hai.

Transform Domain in Computer Vision

– Concept Notes – Part 2 –

Notes by Humma Imran

3. Mashhoor Transforms Kaun Se Hain? (Common Types)

Exam mein aksar poochte hain ke konsi transforms use hoti hain.
Yeh 3 sabse important hain:

Fourier Transform (FT)

- Image ko Spatial Domain (pixels) se Frequency Domain (sine / cosine waves) mein convert karta hai
- Edges aur noise filtering ke liye bohot useful hai

Discrete Cosine Transform (DCT)

- Fourier jaisa transform hai
- Sirf real numbers use karta hai (complex math nahi hoti)
- JPEG Image Compression mein sabse zyada use hota hai

Wavelet Transform

- Zyada advanced transform technique
- Image ki frequency aur location dono ki information store karta hai
- JPEG 2000 format mein use hota hai

4. Spatial Domain vs Transform Domain (Difference)

Feature	Spatial Domain Processing	Transform Domain Processing
• Direct / Indirect	Direct pixels par kaam karta hai	Image pehle transform hoti hai, phir processing hoti hai
• Speed	Chote filters (3×3 mask) ke liye fast	Bare filters ke liye zyada fast aur efficient
• Complexity	Math relatively simple hoti hai (convolution)	Math thori advanced hoti hai (transform equations)
• Best For	Contrast enhancenent, simple sharpening	Image compression, watermarking, heavy noise removal

Exam Tip: Spatial domain simple filtering ke liye use hota hai, jabke transform domain compression aur advanced processing ke liye zyada powerful hota hai.

Transform Domain & Camera Model

— Computer Vision Notes – Part 3 —

Notes by Humna Imran

5. Transform Domain ki Applications (Fayde)

Hum image ko transform domain mein le kar jate hain taake processing zyada powerful aur efficient ho jaye.

Image Compression

- File ka size chota karna,
- Example: 10 MB photo → – 2 MB photo without noticeable quality loss.
- Transform domain mein extra aur unecessary data remove karna easy hota hai.

Watermarking

- Image ke andar hidden copyright ya logo embed kiya jata hai.
- Taake koi image chori na kar ske.

Feature Extraction

- Image mein se important patterns nikalna.
- Example: fingerprints, facial features, object detection.

6. 1. Camera Model Asal Mein Karta Kya Hai? (The Basic Idea)

Humari real duniya 3D space mein hoti hai:

- X, Y, Z (length, width, depth)
- Lekin camera image ko 2D pixels mein convert kar deta hai.

3D World → Camera → 2D Image Plane

Camera Model ek mathematical framework hai jo batata hai:

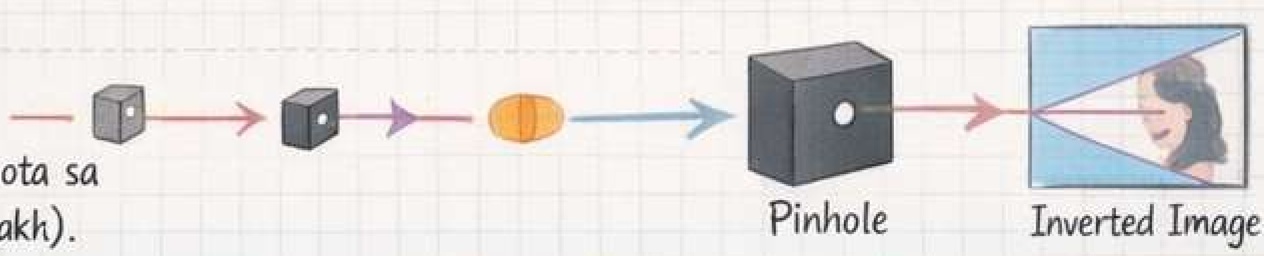
- 3D point (X, Y, Z) ka image par 2D pixel (u, v) kahan appear karega.
- Is process ko Projection kehte hain:

2. Pinhole Camera Model (Exam Ka Sabse Mashhoor Model)

Exams mein aksar Pinhole Camera Model poocha jata hai kyunke yeh sabse basic aur important camera model hai.

Concept:

Ek dark box jiske aage chota sa pinhole (suraakh).



Focal Length (f)

Pinhole aur image plane ke darmiyan jo distance hota hai usay focal length kehte hain.

Camera Model & Coordinate Systems

— Computer Vision Notes – Part 4 —

Notes by Humna Imran.

3. Char Dunyayen (4 Coordinate Systems) – VIP for Exam

Camera setup mein data 4 important stages se guzarta hai.
Exam mein yeh steps sequence mein likhna bohat importanit hai.

World Coordinate System (3D)

- Yeh hamari real duniya hot hai
- Ismein ek universal origin hota hai (0,0,0)
- Example: kamre ka ek kona.

Apple ka position ko sakta hai:
(Xw, Yw, Zw)

Camera Coordinate System (3D)

- Ab hum scene ko camera ki position se dekhte hain
- Camera lens ka center (0,0,0) ban jata hai

Image Plane (2D)

- Yeh camera ka sensor plane hota hai
- Light yahan project hot hai

Coordinates measured in:
(x, y)
units: millimeters

3D World → Camera → Image Plane → Pixel Grid

4. Camera Parameters (The Mathematical Engine)

Camera model ki mathematics parameters par depend karti hai.
Iske 2 main types hote hain:

A. Extrinsic Parameters (Bahar Ki Settings)

- Yeh parameters World Coordinate System ko Camera Coordinate System mein convert karte hain.
- Aasaan zaban mein:
  Camera kahan rakha hai aur kis direction mein dekh raha hai.

Rotation Matrix (R)

- Camera kis angle par ghooma hua hai
- Example: tilt, pan, rotation

Translation Vector (t)

- Camera world origin se kitni door hai
- Camera ki position shift ko represent karta hai.

Origin:
- Top-left corner (0,0)

Exam Tip: Extrinsic parameters camera ki position aur orientation define karte hain.

Camera Parameters & Lens Distortion

– Computer Vision Notes – Part 5 –

– Notes by Humna Imran –

B. Intrinsic Parameters (Andar Ki Settings)

Yeh parameters Camera ki 3D dunya ko screen ke 2D pixels mein convert karte hain.

Aasaan zaban mein:

"Camera ka lens aur sensor kaisa hai?"

→ Intrinsic parameters ko usually matrix **K** se represent kiya jata hai. ←

Focal Length

→ f_x, f_y

- X aur Y direction mein focal length
- Units: pixels

Optical Center / Principal Point

→ C_x, C_y

- Image ka exact center point
- Jahan light directly sensor par hit karti hai

5. The Ultimate Camera Equation

Camera model ki final mathematical relationship is equation se represent hoti hai.

$$x = K \cdot [R \mid t] \cdot X$$

- **X** Asli dunya ka 3D point
- **K** Intrinsic matrix
  Camera coordinates → Pixel coordinates
- **R|t** Extrinsic matrix
  World coordinates → Camera coordinates
- **x** Final 2D pixel point screen par

6. Lens Distortion (Bonus Points Ke Liye)

Asal cameras mein pinhole nahi hota, balkay glass lens hota hai taake zyada light sensor tak pohanch sake.
Lekin lens ki wajah se image ke edges thore distorted ho jate hain.

Radial Distortion

- Seedhi lines image ke edges par curved nazar aati hain
- Example: fisheye camera ya GoPro video

→ Camera calibration ka ek important step distortion correction hota hai.

Exam Tip:

Camera model samgjhne ke liye Extrinsic + Intrinsic parameters aur final camera equation bohat important hain.

Camera Projection Matrix (P)

– Computer Vision Notes – Part 6 –

—— Notes by Humna Imran ——

1. The Ultimate Goal: Camera Matrix (P) 🍌

Computer ko alag alag alag parameters samajh nahi aate.
Isliye hum saari camera settings ko mila kar ek single matrix bana dete hain.

Is matrix ko kente hain:

Camera Projection Matrix (P)

- ✔ Yeh ek 3×4 matrix hota hai,
- ✔ Iska kaam hai 3D world point ko image pixel mein map karna

Agar kisi 3D point ko matrix P se multiply karein,
toh hume pata chal jata hai ke image mein uska pixel al kahan banega.

2. Ingredients (Kaccha Maal) Kya Hai?

Camera Matrix P banane ke liye humein do important parameters chahiye.

Intrinsic Parameters (K)

- ✔ Yeh ek 3×3 matrix hota hai
- ✔ Camera ki internal properties represnt karta

$f_x, f_y \rightarrow$ focal length
$c_x, c_y \rightarrow$ optical center

Extrinsic Parameters [R | t]

- ✔ Yeh ek 3×4 matrix hota hai
- ✔ Camera ki position aur orientation show karta hai

$R \rightarrow$ Rotation matrix (3×3)
$t \rightarrow$ Translation vector (3×1)

3. The Formula: Model Kaise Banta Hai? (VIP for Exam)

$$P = K[R|t]$$

P Final Camera Projection Matrix	**K** Intrinsic Matrix	**[R | t]** Extrinsic Matrix
Size : 3×4	Size : 3×3	Size : 3×4

Exam Tip:

Camera projection matrix $P = K[R|t]$ poora camera model summarize karta hai
aur 3D world points ko 2D image pixels mein convert karta hai.

Homogeneous Coordinates & Camera Matrix Decomposition

— Computer Vision Notes — Part 7 —

Notes by Humma Inran

4. Homogeneous Coordinates (Exam Ka Zaroori Concept)

Jab camera model ki math solve karte hain, ek problem aati hai.

- 3D point mein 3 values hoti hain:

  (,Y, Z,)

- Lekin Camera Matrix P ka size 3 × 4 hota hai.
- Matrix multiplication ke rules ke hisaab se 3 × 4 matrix ko 3 × 1 vector se multiply nahi kar sakte.

Solution: Homogeneous Coordinates

Hum vectors ke end mein extra value "1" add kar dete hain.

3D Point	2D Pixel
(X, Y, Z) becomes → [X, Y, Z, 1]	(u, v) becomes → [u, v, 1]

$$x = P X$$

- - X → 2D pixel point
- - P → Camera matrix
- - X → 3D world point (homogeneous form)

5. Reverse Engineering: Matrix se Parameters Nikalna

Kabhi exam mein Camera Matrix P diya hota hai, aur question hota hai.:

- Intrinsic (K) aur Extrinsic (R, t) parameters nikal kar dikhao.

1. Matrix ko Todna

Camera matrix:

P is 3 × 4

$$\rightarrow \begin{bmatrix} M & Y \\ Y & 1 \end{bmatrix}$$

- Where:
- M → first 3 columns (3 × 3)
- p_4 = last column

2. RQ Decomposition

Matrix M par RQ decomposition apply karte hain.

Result:

$$M = K \times R$$

- Upper triangular matrix → K (Intrinsic matrix)
- Orthogonal matrix → R (Rotation matrix)

3. Translation Vector nikalna

Jab K mil jaye, toh formula use karein:

$$p_4 = K \times t$$

Is se t (translation vector) easily calculate ho jata hai.

Exam Tip: Camera matrix decomposition se K. R, t t parameters nikalna Computer Vision exams ka common question hai.

Camera Calibration

Computer Vision Notes – Part 8

Notes by Humna Imran

1. Camera Calibration Asal Mein Kya Hai? (The Definition)

Pichle topics mein humne dekha ke camera equation banane ke liye humein do parameters chahiye hote hain:

- Intrinsic parameters (K)
- Extrinsic parameters [R|t]

Lekin asal zindagi mein jab aap naya camera khareedte hain, toh uske internal parameters likhe hue nahi hote. Camera ke andar ka:

f_x, f_y aur lens distortion
humein khud calculate karna padta hai.

2. Iski Zaroorat Kyun Parti Hai?

Agar exam mein applications ya importance poochi jaye, toh yeh points likhein:

Metric 3D Measurements

Computer image dekh kar distance measure kar sakta hai.

Example:
- Car camera se kitne meters door hai

Calibration ke bina depth estimate possible nahi hota.

Removing Lens Distortion

Camera lens ki wajah se image edges par curved ho jati hai.

Calibration se humein distortion coefficients milte hain.

Inse hum image ko undistort (seedha) kar sakte hain.

Stereo Vision (3D Vision)

Agar do cameras use ho rahe hon:
- Un dono ka calibrated hona zaroori hai.
- Tabhi system 3D world reconstruction kar sakta hai.

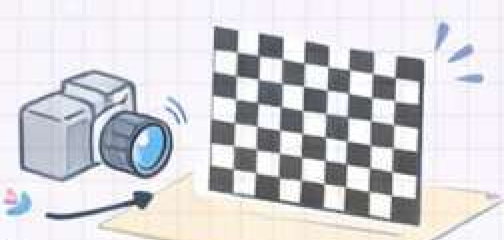
3. The Standard Method: Checkerboard Pattern

(Zhengyou Zhang's Method)

Camera calibration ke bohat se methods hain, lekin industry aur exams mein sabse popular method hai:

Checkerboard (Chessboard) Method

- Hum Checkerboard kyun use karte hain?
  - Board mein black aur white squares hote hain.
  - Squares ke corners detect karna easy hota hai
  - Computer vision algorithms easily corner points detect der lete hain.



Example algorithm: Harris Corner Detector

Is wajah se accurate geometric reference points milte hain.

Exam Tip: Camera calibration ka most common practical method → Checkerboard (Zhang's Method) hai.

Camera Calibration Process
& Reprojection Error

Computer Vision Notes – Part 9

Notes by Humna Imran

4. Calibration Ka Step-by-Step Process (Algorithm Flow)

Exam mein is process ko points ya flowchart ki shakal mein likhna bohat important hai.

Print & Paste

- Ek checkerboard pattern print karein
- Usay flat hardboard par chipka dein
- Square ka size exactly pata hona chaihye

Example: 25 mm square size

→

Take Images

- Camera se checkerboard ki 10–20 images lein

Different:
- angles
- rotations
- distances

→

Find 2D Image Points (u, v)

- Computer algorithm:
- Checkerboard ke corners detect hai
- Har corner ka pixel coordinate (u, v) store karta hai

↓

Define 3D World Points (X, Y, Z)

- Assume:
  Z = 0
- First corner:
  - (0, 0, 0)

Baaki corners square size (25 mm) se calculate hoti hai

5. Reprojection Error (Exam Ka VIP Concept)

Calibration ki accuracy check karne ke liye hum Reprojection Error measure karte hain.

Concept

- Computer jo parameters calculate karta hai, unko use karke:

3D points ko dobara image plane par project karta hai.

Error:

- Difference between:
- Original detected 2D corner point
- Reprojected 2D point

In dono ke darmiyan jo distance hota hai usay kehte hain:

Reprojection Error

Goal:

- Calibration tab successful hoti hai jab:

Reprojection Error < 1.0 pixel

Exam Tip: Camera calibration ki accuracy evaluate karne ka best metric → Reprojection Error hai

Levenberg–Marquardt optimization.